Solution Architecture

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Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Phase 3: Project Design Phase - Solution Architecture

1. Architectural Layer Breakdown

The solution follows ServiceNow's standard 3-tier enterprise architecture:

Architecture Layer	ServiceNow Component	Role & Functionality
Presentation Layer	Service Catalog Portal (sc_cat_item)	Displays Car Rental Request item, renders 7 variables & choices (Mini, SUV, Sedan).
Process & Workflow Layer	Flow Designer (Car Rental Req flow)	Executes approval rules, pauses for approval response, generates sc_task, fetches catalog variables, dispatches emails.
Business Logic & Data Layer	Business Rule + GlideRecord (sc_req_item, sc_task, incident)	Executes GlideRecord script to create Incident ticket when car state = 3 (Unavailable). Stores audit history.

2. Data Entity Relationship Model

• Request (sc_request) [1] ----> [N] Requested Item (sc_req_item)

• Requested Item (sc_req_item) [1] ----> [N] Catalog Task (sc_task)

• Requested Item (sc_req_item) [1] ----> [1] Incident (incident) [Created via Business Rule on State=3]